Answers: Introduction to set theory

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Summary

Answers to questions relating to the guide on introduction to set theory.

These are the answers to Questions: Introduction to set theory.

Please attempt the questions before reading these answers!

Q1

1.1. This set is finite.

1.2. This set is countably infinite.

1.3. This set is finite.

1.4. This set is uncountable.

1.5. This set is uncountable.

1.6. This set is countably infinite.

Q2

2.1. This is TRUE.

2.2. This is FALSE, as $2 \notin \{1, 3, 5, 6, 7, 16, 18, 20\}$.

2.3. This is FALSE, as $\frac{1}{3}, \frac{41}{2} \notin \mathbb{Z}$.

2.4. This is TRUE.

2.5. This is FALSE, as $(1.6, 5]$ is open on the left, so $1.6 \notin (1.6, 5]$.

2.6. This is TRUE.

2.7. This is FALSE, as 15 is not even, so $15 \notin \{x : \frac{x}{2} \in \mathbb{Z}\}$.

2.8. This is TRUE, as $6, 18, 24$ and 27 are all multiples of 3.

Q3

3.1.

$$\mathcal{P}(A) = \begin{array}{l} \{\ \emptyset, \\ \{1\},\ \{2\},\ \{3\}, \\ \{1,2\},\ \{1,3\},\ \{2,3\}, \\ \{1,2,3\}\} \end{array}$$

3.2.

$$\mathcal{P}(B) = \begin{array}{l} \{\ \emptyset, \\ \{\frac{1}{2}\},\ \{\frac{1}{3}\},\ \{\frac{1}{6}\}, \\ \{\frac{1}{2},\frac{1}{3}\},\ \{\frac{1}{2},\frac{1}{6}\},\ \{\frac{1}{3},\frac{1}{6}\}, \\ \{\frac{1}{2},\frac{1}{3},\frac{1}{6}\}\} \end{array}$$

3.3.

$$\mathcal{P}(C) = \begin{array}{l} \{\ \emptyset, \\ \{1\},\ \{3\},\ \{6\},\{9\}, \\ \{1,3\},\ \{1,6\},\ \{1,9\},\{3,6\},\{3,9\},\{6,9\}, \\ \{1,3,6\},\{1,3,9\},\{1,6,9\},\{3,6,9\}, \\ \{1,3,6,9\}\} \end{array}$$

3.4.

$$\mathcal{P}(D) = \begin{array}{l} \{\ \emptyset, \\ \{\emptyset\},\ \{4\}, \\ \{\emptyset,4\}\} \end{array}$$

3.5.

$$\mathcal{P}(E) = \begin{array}{l} \{\ \emptyset, \\ \{4.5\},\ \{8\},\ \{9\},\{10\}, \\ \{4.5,8\},\ \{4.5,9\},\ \{4.5,10\},\{8,9\},\{8,10\},\{9,10\}, \\ \{4.5,8,9\},\{4.5,8,10\},\{4.5,9,10\},\{8,9,10\}, \\ \{4.5,8,9,10\}\} \end{array}$$

3.6.

$$\mathcal{P}(F) = \begin{array}{l} \{\ \emptyset, \\ \{\sqrt{2}\},\ \{5\},\ \{\frac{16}{3}\}, \{18.2\}, \\ \{\sqrt{2}, 5\},\ \{\sqrt{2}, \frac{16}{3}\},\ \{\sqrt{2}, 18.2\}, \{5, \frac{16}{3}\}, \{5, 18.2\}, \{\frac{16}{3}, 18.2\}, \\ \{\sqrt{2}, 5, \frac{16}{3}\}, \{\sqrt{2}, 5, 18.2\}, \{\sqrt{2}, \frac{16}{3}, 18.2\}, \{5, \frac{16}{3}, 18.2\}, \\ \{\sqrt{2}, 5, \frac{16}{3}, 18.2\}\} \end{array}$$

Q4

4.1. $|\mathcal{P}(A)| = 2^{|\mathcal{P}(A)|} = 2^3 = 8$

4.2. $|\mathcal{P}(B)| = 2^{|\mathcal{P}(B)|} = 2^3 = 8$

4.3. $|\mathcal{P}(C)| = 2^{|\mathcal{P}(C)|} = 2^4 = 16$

4.4. $|\mathcal{P}(D)| = 2^{|\mathcal{P}(D)|} = 2^2 = 4$

4.5. $|\mathcal{P}(E)| = 2^{|\mathcal{P}(E)|} = 2^4 = 16$

4.6. $|\mathcal{P}(F)| = 2^{|\mathcal{P}(F)|} = 2^4 = 16$

Version history and licensing

v1.0: initial version created 09/24 by tdhc.